inertiaforge: Auditing Contact Dynamics and the Limits of Gradient Agreement

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Abstract

We develop a self-contained three-dimensional rigid-body engine and use it to test whether agreement across parameter perturbations, timesteps and contact regimes makes a numerical sensitivity trustworthy. The proposed empirical controller changes both simulation effort and optimization updates. It fails its preregistered accuracy criterion: on 19 smooth held-out sphere encounters, it accepts 18 derivatives, but only three accepted derivatives meet the declared error tolerance. A separate event-time implementation of established mechanics passes all 19 supported smooth cases and refuses a grazing case. We explain the failure with a uniform-convergence counterexample and independently derived impact sensitivities. The engine also supports rotation, Coulomb friction, simultaneous contact manifolds and constrained mechanisms. Independent reconstruction verifies physical checks, external CPU comparisons and a measured-data experiment: calibration on one GAUGE bouncing-ball trial transfers to three untouched trials with 5.474 mm pooled vertical error, compared with 47.452 mm for authored parameters. Strong MuJoCo and Newton controls remove several apparent advantages. Optional compilation yields workload-dependent gains, not a universal speedup. The engine supports reproducible tests of contact dynamics and gradient reliability within the stated model.

1 Problem and scope

A contact-rich trajectory can look plausible while its derivative describes the wrong limiting physical model. An optimizer may also benefit from a direction that is not that model's derivative. These questions require separate evidence. We distinguish (i) fidelity of a forward trajectory within a declared contact law, (ii) accuracy of a sensitivity of that law, and (iii) usefulness of an optimization update.

Our initial hypothesis was that combining contact-regime traces with parameter and timestep refinement could allocate computation more effectively than simple finite differences. Before evaluating the candidate, we fixed two acceptance targets: at least 95% of accepted smooth-case derivatives must have scaled error at most 5%, at coverage of at least 70%; and a downstream identification advantage must reduce complete cost by at least 25% against the strongest applicable simple comparator on two contact families. The complete contract is retained with the code. Neither target is relaxed in response to results.

The supported engine domain is deliberately bounded: rigid spheres and oriented boxes, infinite static planes, gravity and applied world forces/torques, isotropic Coulomb friction, Newton restitution, and ball-joint constraints. Supported collision pairs are sphere–sphere, sphere–plane, box–plane and sphere–box. Multiple vertices supply a box–plane manifold. Box–box collision is explicitly rejected. There is no general mesh, deformable, fluid or complex robot capability. Serial independent worlds enable repeatable experiments without implying GPU throughput.

This report contributes an inspectable implementation and a concrete negative experimental finding. The underlying mechanics, continuous collision detection, event-time differentiation, adaptive numerical checks and derivative-free search are established techniques. Our proposed combination does not acquire novelty merely because it has a new implementation.

2 Prior work and competing routes

Continuous event-time sensitivity is well established. The saltation matrix accounts for the change in event time as well as the reset map [6]. Zhong et al. show why a changing contact normal requires collision response at the time of impact, not only a position correction [12]. These works motivate a strong event-time comparator rather than a deliberately weak fixed-grid baseline.

DiffMJX couples adaptive integration with penalty-based simulation; its Contacts From Distance mechanism changes the backward pass while preserving the forward simulation [9]. Bundled Contact Gradients advances perturbed branches and averages their states before continuing [1]; this changes the forward construction and must not be conflated with a backward-only modification. Randomized gradient bundling also predates that work [11]. Adaptive finite-difference interval selection balances truncation error against function-evaluation error [10]. Therefore, shrinking a perturbation or switching an estimator is not by itself a new research contribution.

We considered three routes. The selected controller jointly adjusts perturbation and temporal resolution. A second route would bound a local active-set radius for a fixed contact linear complementarity problem; it requires nonsingularity and does not directly handle changing geometry or friction. A third route would control the discrepancy between smoothed and hard coupled contacts; it requires justified error propagation rather than treating a solver residual as a trajectory bound. Source inspection did not establish that any route was novel. The latter two remain proposals, not implemented findings.

The engine audit follows actual Newton, Warp, MuJoCo, MuJoCo Warp and Genesis solver/test paths [7, 8, 3, 4, 2]. For example, Newton XPBD offers a relevant independent rigid-contact comparator, while its semi-implicit penalty solver has different friction semantics. MuJoCo's compliant contact parameters do not provide a direct universal mapping to a hard restitution coefficient. Public DiffMJX components were inspected but not locally benchmarked; BCG's project page labeled code forthcoming. No comparison in this report claims to reproduce those complete systems.

3 Rigid-body formulation and implementation

3.1 State and free motion

For body i, state is $(x_i, q_i, v_i, \omega_i)$, with scalar-first unit quaternion q_i, world angular velocity ω_i, mass m_i and body-frame diagonal inertia I_i^B. Static bodies have zero inverse mass. The world inertia is

$$I_i(q_i) = R(q_i) I_i^B R(q_i)^T. \tag{1}$$

With constant applied force over a step h, translation uses

$$v^* = v + h(g + f/m), \qquad x^* = x + hv + \tfrac{1}{2}h^2(g + f/m). \tag{2}$$

This is exact for constant-acceleration free flight. It is an integration choice, not a contact-method advance. Rotation is advanced through world angular momentum and midpoint orientation; this

incorporates the effect represented by the gyroscopic term $\omega \times I\omega$. Isotropic sphere inertia admits a simpler update. Quaternions are normalized. Free-flight and torque-free invariant checks test this path separately from contact.

3.2 Impulse contact and constrained motion

Contact normals point from body b toward body a. At a contact point with offsets r_a, r_b, the relative point velocity is

$$u = (v_a + \omega_a \times r_a) - (v_b + \omega_b \times r_b). \tag{3}$$

The effective inverse mass for an unconstrained contact is

$$K = (m_a^{-1} + m_b^{-1})I - [r_a]_\times I_a^{-1}[r_a]_\times - [r_b]_\times I_b^{-1}[r_b]_\times. \tag{4}$$

We accumulate impulses rather than repeatedly applying an unconstrained impact. Normal impulses are projected onto $\lambda_n \geq 0$; tangential impulses are projected onto the disk $\|\lambda_t\|_2 \leq \mu\lambda_n$. Pair friction uses the geometric mean and restitution the larger material coefficient. Restitution is suppressed below a configured incoming-speed threshold to avoid indefinitely bouncing near rest. These conventions are explicit parts of the model.

Ball joints impose coincident anchors. When J is the joint velocity Jacobian, the constraint-compatible inverse mass used for contact is the established Schur projection

$$W = M^{-1} - M^{-1}J^T(JM^{-1}J^T)^{-1}JM^{-1}. \tag{5}$$

This prevents contact impulses from immediately violating the eliminated joint constraints. It does not guarantee uniqueness or convergence for every simultaneous frictional impact. A high-friction constrained impact is retained as a failing case, with nonconvergence visible in diagnostics.

Separate nonlinear position projection reduces penetration and anchor drift. It changes configuration without directly changing velocity. This can change potential energy and, for anisotropic bodies at fixed world angular velocity, rotational kinetic energy. The engine records this energy change. Small post-projection penetration is not evidence that the trajectory, event time or derivative is correct.

3.3 Diagnostics and interfaces

A scene, reusable state, control and step result form the headless API. Step results report contact points, normals, impulses, friction limits, regime labels, energy, overlap and constraint residuals. Normal projected residuals have impulse units; joint velocity residuals have velocity units. Their tolerances are numerical stopping conditions, not physical error bounds.

Every substep's convergence is aggregated. Independent review found that the initial wrapper retained only the last substep's status: a ten-substep stack could report success after three earlier failures. The corrected wrapper reports failure, the failed-substep count and maxima, while preserving final-substep diagnostics separately. The original failure and its recheck remain in the evidence directory.

The reference implementation uses double-precision NumPy. An optional Numba backend compiles contact sweeps only. Collision detection, state semantics and diagnostics remain shared. Saved states preserve warm-start information; input states are not mutated. Deterministic configurations and full JSON experiments support replay.

4 Sensitivity controller and why it can fail

Let $J(\theta)$ be an observable of the declared physical flow and $J_h(\theta)$ its discretized approximation. The derivative of the implemented program, DJ_h, is not generally a reliable approximation to DJ. For perturbation ϵ, define

$$C_{h,\epsilon} = \frac{J_h(\theta+\epsilon) - J_h(\theta-\epsilon)}{2\epsilon}, \tag{6}$$

$$D^+_{h,\epsilon} = \frac{J_h(\theta+\epsilon) - J_h(\theta)}{\epsilon}, \qquad D^-_{h,\epsilon} = \frac{J_h(\theta) - J_h(\theta-\epsilon)}{\epsilon}. \tag{7}$$

The candidate compares central secants at ϵ and $\epsilon/2$, directional disagreement, temporal disagreement at h and $h/2$, and forward-observable disagreement. It compresses only consecutive repeated contact signatures; contact identities and their ordering remain intact. Passing these empirical checks accepts the derivative. Otherwise it refines within a fixed budget, then abstains. In identification, abstention switches the update to a bounded directional poll. Thus the mechanism changes computation and optimization rather than merely displaying a warning.

The frozen tolerances are $0.002 + 0.02\max(|C_{h,\epsilon}|, |C_{h,\epsilon/2}|)$ for derivative comparisons, twice this value for one-sided disagreement, and 10^{-4} for forward agreement. The held-out study permits four refinement levels, starts at $h = 0.008\,\mathrm{s}$ and $\epsilon = 0.001\,\mathrm{m}$, and compares an unconditional $h = 0.0005\,\mathrm{s}$ policy. These values define a tested algorithm, not a universal tolerance prescription.

If a valid observable-error bound $|J_h(\theta) - J(\theta)| \le \delta(\theta)$ existed, its contribution to central-difference error would be bounded by

$$\frac{\delta(\theta+\epsilon) + \delta(\theta-\epsilon)}{2\epsilon}. \tag{8}$$

A contact residual alone does not supply δ: conditioning and propagation through the trajectory are missing. Nested-grid differences are only empirical substitutes.

Proposition 1. *Uniform forward convergence and agreement of nested-grid derivatives do not establish convergence to the derivative of the limiting observable.*

Proof. Set $J_h(\theta) = \theta + h\sin(\theta/h)$. Then $|J_h(\theta) - \theta| \le h$, so J_h converges uniformly to $J(\theta) = \theta$. Nevertheless $J_h'(0) = 2$ for every h, while $J'(0) = 1$. When ϵ/h is small, central and both one-sided secants at h and $h/2$ approach 2. Forward differences also become small. The implemented empirical gate accepts a value approximately 2 for the retained test. It is not a certificate. □

This elementary counterexample is used as an adversarial test, not claimed as a new mathematical theorem. The physical study below shows the same practical distinction between a converged discrete branch and the desired impact sensitivity.

5 Independent impact reference

For two frictionless spheres, let initial relative position be r, relative velocity v, and combined radius R. With equal gravitational acceleration, relative free motion is linear. An isolated time of impact τ satisfies

$$\|r + \tau v\|^2 = R^2. \tag{9}$$

We choose the earliest approaching root, compute the normal at that root and apply the standard restitution impulse. The implementation rejects simultaneous impacts, initial penetration, resting

contacts, friction and joints. It is a useful restricted baseline, not a replacement for the full discrete engine.

For a parameter perturbing r, differentiation of the guard gives

$$\tau' = -\frac{n^T r'}{n^T v}, \qquad n' = \frac{r' + v\tau'}{R}, \qquad n = \frac{r + \tau v}{R}. \tag{10}$$

The normal and remaining post-impact flight time both depend on the parameter. Differentiating only the reset at a fixed step boundary omits terms. The independent reference computes these expressions separately from the engine; a reviewer reconstructed them again using a scalar chord-length formulation.

At grazing, $n^T v$ loses transversality. One must inspect the particular observable instead of declaring a scalar derivative. For the retained $R = 0.5\,\mathrm{m}$, speed $3\,\mathrm{m/s}$, restitution 0.6, mass ratio 1.5, horizon $0.8\,\mathrm{s}$ case, the final striker x-position has left and right offset derivatives 5.376 and 0, respectively. The scalar derivative is undefined. All empirical methods in the held-out run refuse that boundary case; it is excluded from smooth accuracy denominators and reported separately.

6 Experiments and evidence

6.1 Physics verification

The verification suite uses analytical free fall/projectiles, equal and unequal-mass impacts, momentum and energy checks, friction-cone limits, constrained oscillation and timestep refinement, stacks, torque-free rotation, replay and independent worlds. Slip-to-roll behavior is also checked independently against the analytical contact impulse and conserved quantities. Invalid quaternions, nonfinite inputs and unsupported geometry fail explicitly. Tests of the numerical controller include a smooth observable, a nondifferentiable absolute-value kink and a false-convergence counterexample.

The contact pendulum is a meaningful driven constrained system, but not a claim of arbitrary contact-robot robustness. A high-friction configuration causes the coupled impulse iteration to fail its declared tolerance. The engine reports the failure, and the laboratory displays it persistently. Raising the iteration cap is measured separately rather than silently relabeling all states as converged.

6.2 Held-out contact derivatives

We reserve seeds 4100–4199 separately from development seeds 100–199. The executed derivative set comprises 16 randomized oblique encounters plus four boundary/miss cases, with one exact grazing case outside the smooth domain. Speed, mass ratio, restitution and horizon vary; this changes the collision phase relative to the timestep. Three runs per method/case measure timing repeatability, not independent physical samples. Six methods give 360 recorded outputs.

The candidate passes the coverage target but decisively fails accuracy. Removing temporal refinement does not rescue reliability; its accepted accuracy remains only 31.6%. Unconditionally reducing the timestep can still stabilize the wrong discrete derivative. The event-time baseline succeeds in this narrow domain, as existing time-of-impact reasoning predicts. We do not generalize that result to friction, stacks or simultaneous impacts.

Removing contact-signature agreement changes none of the 20 unique numerical outcomes in the retained ablation; contact metadata adds no demonstrated benefit in this family. The main conclusion does not depend on declaring this small controller the strongest adaptive differentiation method: it

Method	Smooth coverage	Accurate / accepted	Median cost (s)	Median steps
Parameter-only refinement	19/19	6/19	0.097	525
Coupled controller	18/19	3/18	0.341	1223.5
Fine timestep + refinement	16/19	2/16	2.727	12372.5
Isolated event-time secants	19/19	19/19	0.00269	5

Table 1: Smooth held-out configurations. Accuracy means absolute derivative error divided by $\max(1, |DJ|)$ is at most 0.05. Times and step medians include the full population, including the grazing refusal, and all requested diagnostic rollouts. A closed-form event derivative is an additional model-specific reference.

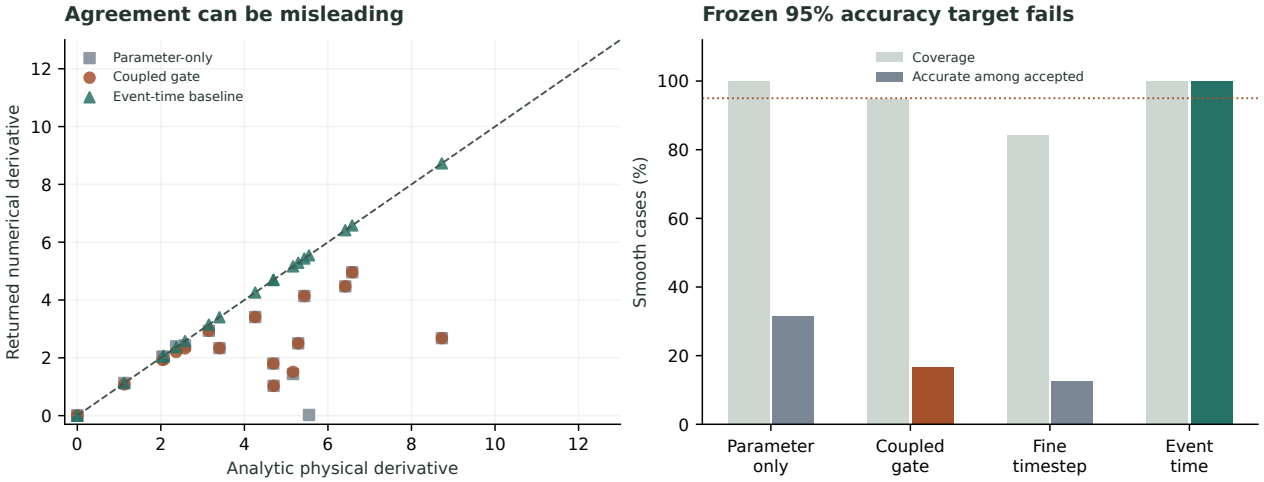


Figure 1: Actual held-out derivatives and acceptance outcomes. The identity line is the independent physical reference. Three timing repetitions are collapsed to unique cases in the scatter. High coverage does not imply trustworthy acceptance.

fails its own frozen target and is unsuitable for certified-gradient claims. We have not executed every modern differentiable simulator or proven that the broader family of adaptive methods must fail.

6.3 Measured GAUGE transfer

GAUGE provides real 30 Hz bouncing-ball trajectories [5]. We verified public access and inspected the actual USD geometry rather than assuming a README was a calibrated model. The ball is approximately spherical with radius 0.03000003 m. The authored material has restitution about 0.6 and zero friction; mass is not authored. Vertical sphere–plane motion is mass-independent in this model, so using a nominal mass is not mass identification.

Trial 1 calibrates restitution and one effective contact-center height. Trial 22 was inspected during development and is not held out. Before downloading trials 2, 3 and 4, we froze a common initialization rule: use frames 4–8 for a ballistic initial height/velocity fit, start from frame 8, and score frames 9 onward. No held-out recropping, time-shift fitting or material fitting is allowed. All three held-out initialization windows pass the fixed check; no trial is excluded.

Calibration gives restitution 0.727029 and effective contact-center height 0.038312 m. The latter is an observation/coordinate nuisance parameter, not independently verified floor geometry. The analytic calibration is separate from the engine, which predicts at 240 and 960 Hz. Across 191 held-out

frames, authored parameters give 47.452 mm pooled vertical RMSE in the 960 Hz engine; calibrated parameters give 5.474 mm. The independent analytic predictor gives 5.287 mm after calibration. At 240 Hz, calibrated engine error rises to 8.985 mm.

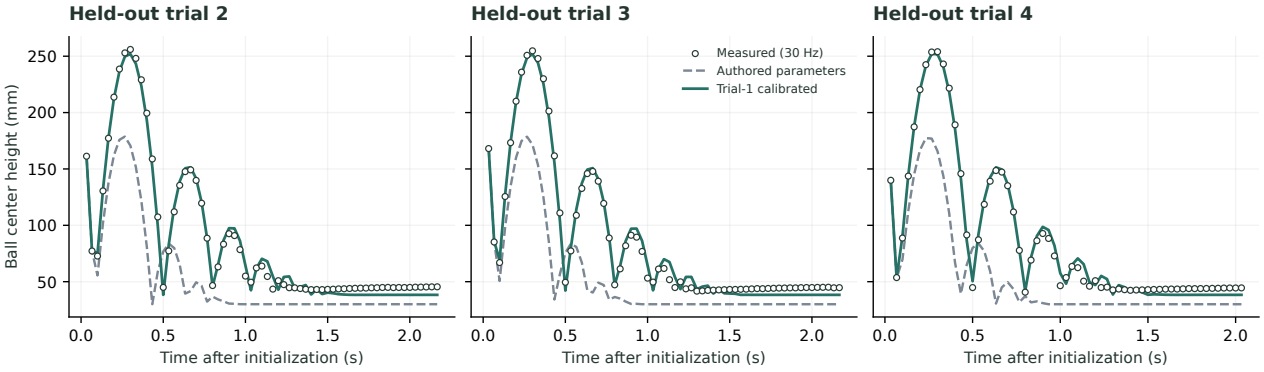


Figure 2: Three untouched measured trajectories with fixed initialization. Markers are source observations; curves are actual engine predictions. Only trial 1 selects restitution and the shared observation-height offset. The result is a narrow vertical-transfer experiment, not an official GAUGE benchmark reproduction.

An independent reconstruction refits calibration using a geometric impact-time schedule, reproducing restitution within 1.6×10^{-12} and height within 2.1×10^{-13} m. That validates computation, not measurement truth. Sensor uncertainty, coordinate convention, unmodeled compliance and only three held-out trials limit the physical conclusion. This experiment does not demonstrate that the rejected sensitivity controller is useful. The original calibration records reported function-evaluation counts but not complete residual-call cost; no total identification-speed claim is inferred from those counts.

6.4 External engines and strong controls

The CPU study runs 42 configurations and 126 warmed trajectories. The common sphere has radius 0.25 m, mass 1 kg, gravity 9.81 m/s^2; drop height is 1.5 m and duration 0.8 s. Timesteps are 8, 4 and 2 ms. Tested packages are MuJoCo 3.17.0, Newton 1.7.0 and Warp 1.19.0. Newton uses CPU float32; inertiaforge and native MuJoCo use double precision. Equivalence limits are explicit.

Initial comparisons were challenged with stronger simple controls. MuJoCo RK4 matches inertiaforge's constant-acceleration free-fall accuracy, removing the apparent accuracy advantage over semi-implicit integration. Two Newton XPBD iterations produce the same one-contact trajectories as forty, with much lower cost. A 0.05 m collision-search gap and zero physical margin removes Newton's apparent sampled-penetration disadvantage. These controls remain in the final evidence rather than only the original unfavorable defaults.

At 2 ms, the retained initial inertiaforge bounce error is 1.072 mm versus 2.615 mm for the Newton two-iteration/search-gap control. This is an integration/model-policy result on one simple scene, not a new contact algorithm. MuJoCo's uncalibrated compliant bounce has different dynamics and cannot be ranked as an equivalent hard-restitution model. Timings include Python loops, collision dispatch and readback; raw records separate setup, warmup/possible compilation, stepping and readback. Host activity is uncontrolled, so three repeats provide observed ranges rather than a broad statistical guarantee.

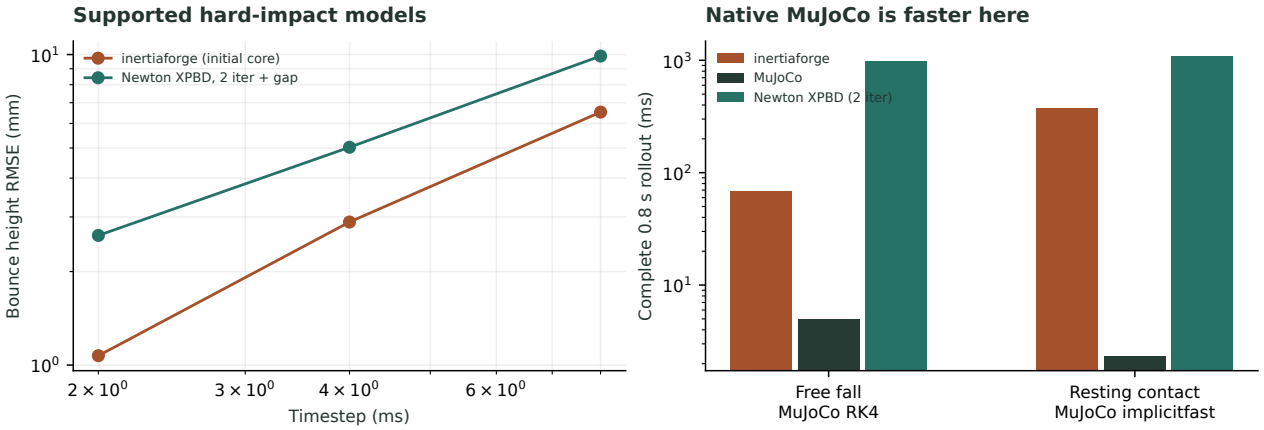


Figure 3: Independent hard-impact error comparison and complete CPU rollout times. Inertiaforge results in this figure use the retained initial source snapshot. MuJoCo bounce is omitted from the hard-impact ranking because its uncalibrated compliant model is different. Native MuJoCo is substantially faster on these one-body workloads.

6.5 Execution cost and independent worlds

Profiling identified repeated small-array and cross-product work in the reference path. Caching inertia, avoiding unnecessary contact-free work and compiling contact sweeps were motivated by these measurements. NumPy remains the reference; compilation is opt-in. The complete accepted-stack benchmark uses 160 maximum sweeps and reports zero failed steps. A cheaper 40-sweep benchmark with three failed steps is preserved separately and is not used as evidence of accepted-solve performance.

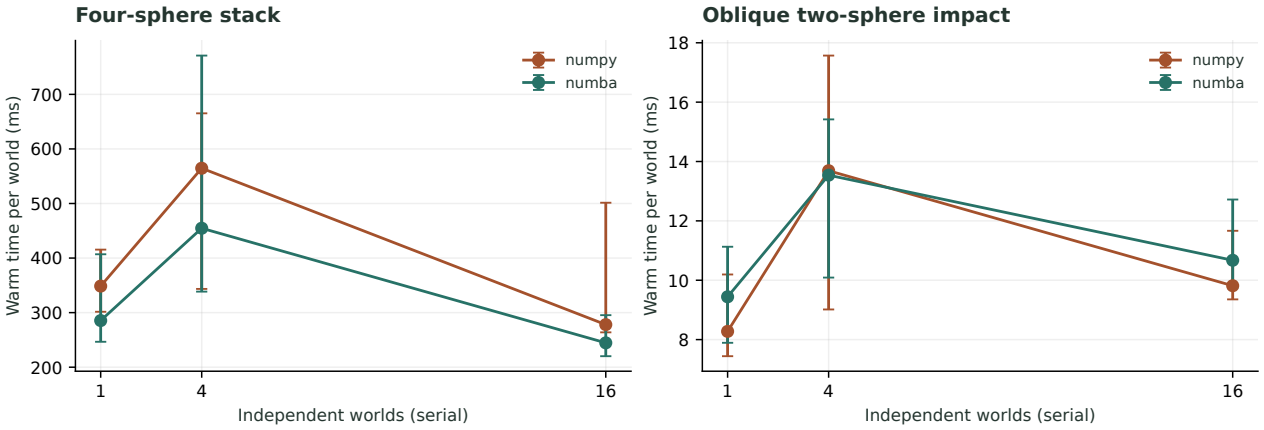


Figure 4: Seven shuffled warmed repetitions at 1, 4 and 16 serial independent worlds. Error bars are observed minima/maxima. Contact compilation helps the stack modestly but does not consistently help sparse oblique impacts. This is neither GPU scaling nor a parallel-world throughput claim.

The accepted stack shows approximately 1.14–1.24 times warmed NumPy/Numba speed ratios across the tested world counts. Oblique ratios range approximately 0.88–1.01, so compilation is not universally beneficial. First-call compilation/loading cost is retained. Separate fresh-process measurements give approximately 34 MB peak resident memory for NumPy and 121 MB for Numba, including im-

ported runtime/compiler dependencies. Python-traced peaks are much smaller and exclude untraced native allocations; they are not total process memory. Source bytes and hashes pair each timing artifact with the version actually executed; later input-validation additions are not silently assigned earlier timings.

7 Identification and the intervention test

The downstream study uses synthetic analytic observations for lateral-offset and restitution identification. It compares fixed finite-difference updates, the coupled gate with an actual directional-search fallback, a fine-resolution scalar search and an event-time scalar search. Physical parameter error and held-out analytic prediction are separated from replay through the discrete solver: accurate physical parameters may still produce biased discrete predictions, while fitted parameters may compensate discretization error.

Every actual observable call, simulation step, refinement and rejected update is counted. Fits and held-out validation costs are separate and also summed. Unlike a solver's reported `nfev`, these counters include finite-difference work. A fixed set of cases and three timing repetitions are retained. A common loss target applies to all optimizers. During review, the floor's default position was found inconsistent with the analytic restitution model and corrected before restitution cases ran; incomplete offset-only results and the earlier protocol remain archived. The completed results follow.

The final study records 60 runs across five cases (three offsets and two restitutions), with 2,730 actual simulator calls including validation and 360 separately counted analytic-reference evaluations. All final runs use one immutable runtime snapshot. The candidate changes actions but fails the two-family cost target. Its median fitting time is 5.42 times fixed finite differences for offset identification and 3.01 times for restitution. It is 4.29 times the fine scalar-search cost for offset, and 0.715 times for restitution: this raw fitting-time ratio is not a matched-accuracy advantage, and it does not satisfy the frozen two-family target.

Family	Method	Median fit (s)	Median calls	Held-out physical MAE (mm)
Offset	Fixed finite differences	1.029	29	7.134
	Coupled gate	5.572	118	5.686
	Fine scalar search	1.298	23	7.163
	Event-time scalar search	0.0084	23	0.0173
Restitution	Fixed finite differences	0.317	13	2.475
	Coupled gate	0.955	28	2.475
	Fine scalar search	1.337	21	2.339
	Event-time scalar search	0.0072	21	below 10^{-9}

Table 2: Synthetic inverse problems. Physical error uses independent analytic predictions at changed held-out conditions. The raw record also reports fine-core replay error, parameter error, fitting plus validation costs, and all actual calls. Medians aggregate three timing repetitions per frozen case.

The isolated event-time baseline is most accurate on these supported physical models. Replaying its accurate parameters through the discrete solver still yields millimetre-scale errors. This distinction matters: minimizing a discrete loss can identify a parameter that compensates for the discretization instead of the physical parameter generating the observation. The five noiseless one-parameter cases do not establish global identifiability in a coupled multiparameter system.

Complete fit-plus-validation median cost ratios against fixed finite differences are 2.535 for offset

and 1.769 for restitution. These costs also fail to establish the requested advantage. An independent reviewer reconciled all 2,730 simulator calls, 721,560 discrete steps and 360 analytic-reference evaluations, and reproduced physical-reference predictions to 1.33×10^{-15} m.

8 Laboratory, reproducibility and limitations

The local laboratory runs the same Python engine used headlessly. It exposes parameter units, orbitable three-dimensional playback, trajectory scrubbing, contact normals and impulses, energy and constraint plots, separate timestep-comparison cost and JSON replay/export. Changing a parameter marks the displayed result stale. Numerical nonconvergence is a persistent visible warning; it does not erase the result or pretend that a plausible animation is accurate. Three.js renders supplied states and does not advance physics.

The repository contains the core, examples, tests, source audit, frozen contract, raw outputs, independent reconstruction, figure generator, editable LaTeX and compiled PDF. A minimal wheel requires NumPy; research and compiled execution dependencies are optional. No paid API, credentials, remote compute or GPU measurements were used. CPU architecture and available logical processors are recorded; denied hardware-detail queries are left unverified.

Important limits remain. Discrete collision detection can tunnel; point contacts and instantaneous restitution do not capture finite compliance. Multiple simultaneous frictional impacts need not have a unique resolution. The event backend has a much smaller domain than the main solver. Sensitivities are numerical secants, not a full automatic-differentiation system. The rejected gate must not be used as a certificate. Real-data evidence is one calibrated vertical model and three test trajectories. The external engine workload is small and host timing noise is substantial. DiffMJX, BCG and high-dimensional control are not reproduced. We have not established a novel research mechanism with the requested practical advantage.

9 Conclusion

A real, reproducible contact engine makes a failed sensitivity hypothesis inspectable. The proposed agreement controller cannot reliably distinguish a stable discrete derivative from the derivative of the intended impact model. Established event-time mechanics solves the supported isolated-impact comparison, while carefully initialized measured trajectories support limited parameter-transfer evidence. Strong baselines and independent reconstruction prevent these useful results from becoming unsupported claims of novelty, certification or universal speed. The unresolved research problem is a consequential, cost-effective reliability mechanism for coupled frictional and changing-contact dynamics beyond the isolated-event domain.

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